

MedSphere – Unified Hospital Information Management System

MCA Semester 3 Final Project Report

1. Introduction

MedSphere is a Java/Spring Boot hospital information management application developed as an MCA Semester 3 mini project.

2. Problem Statement

Hospital information workflows connect patients, doctors, departments, appointments and clinical records. MedSphere provides a centralized academic system for these core workflows.

3. Objectives

Centralized patient information; role-based access; department and doctor management; appointment scheduling; consultation and prescription recording; medical history; validation; persistence; secure authentication.

4. Functional Modules

Authentication; Patient Management; Department Management; Doctor Management; Appointment Management; Doctor Consultation; Prescription Management; Medical History; Dashboard and responsive UI.

5. Technology

Java 21; Spring Boot 3.5.5; Spring MVC; Spring Security; Spring Data JPA; Hibernate; Thymeleaf; Bootstrap 5.3.3; JavaScript; MariaDB/MySQL; Maven Wrapper; XAMPP.

6. Architecture

Browser/UI → Controller → Service → Repository → JPA/Hibernate → MariaDB/MySQL.

7. Security

Database-backed authentication, BCrypt password hashing, CSRF protection and role-based authorization.

8. Database

Main entities are Role, User, Department, Doctor, Patient, Appointment, Consultation, Prescription and MedicalHistory. Patient codes use MSP-YYYY-000001.

9. Core Workflows

Patient registration → patient code → profile/history.

Appointment creation → conflict validation → SCHEDULED.

Doctor consultation → save → COMPLETED.

Consultation → prescription.

Admin → doctor account and profile.

10. Testing

Manual functional testing covered authentication, authorization, CRUD workflows, validation, appointment conflicts, consultation/prescription flow and persistence after restart.

11. Limitations

Billing, pharmacy inventory, laboratory management, wards/beds, advanced notifications and analytics are not implemented in the Semester 3 version.

12. Future Scope

Semester 4 can expand EMR, laboratory/radiology, pharmacy, billing, admission/discharge, bed management, notifications, analytics, backup/recovery and multi-building/multi-branch architecture.

13. Conclusion

MedSphere demonstrates an end-to-end Java/Spring Boot hospital information workflow with secure role-based access, relational persistence and layered architecture.